

Multilingual Emotion Detection

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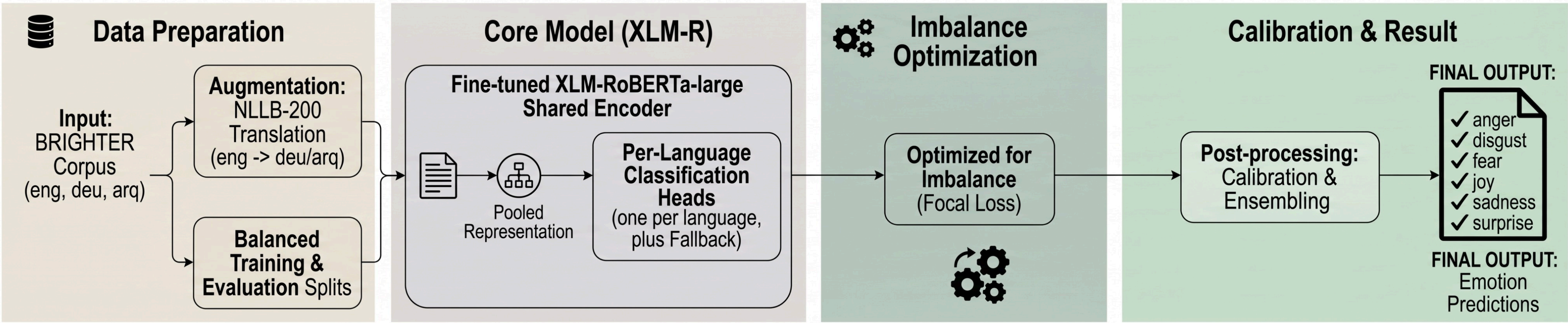


Introduction

- Detect emotions in multilingual short texts from the BRIGHTER corpus.
- Predict multiple emotions per text: anger, fear, joy, sadness, surprise, and disgust.
- Focus on main three languages: English, German, and Algerian Arabic.
- Test fallback-head performance on Persian, Spanish, Hausa, and Amharic using 50 translated sentences.
- Address low-resource and imbalanced labels, especially rare emotions in German.

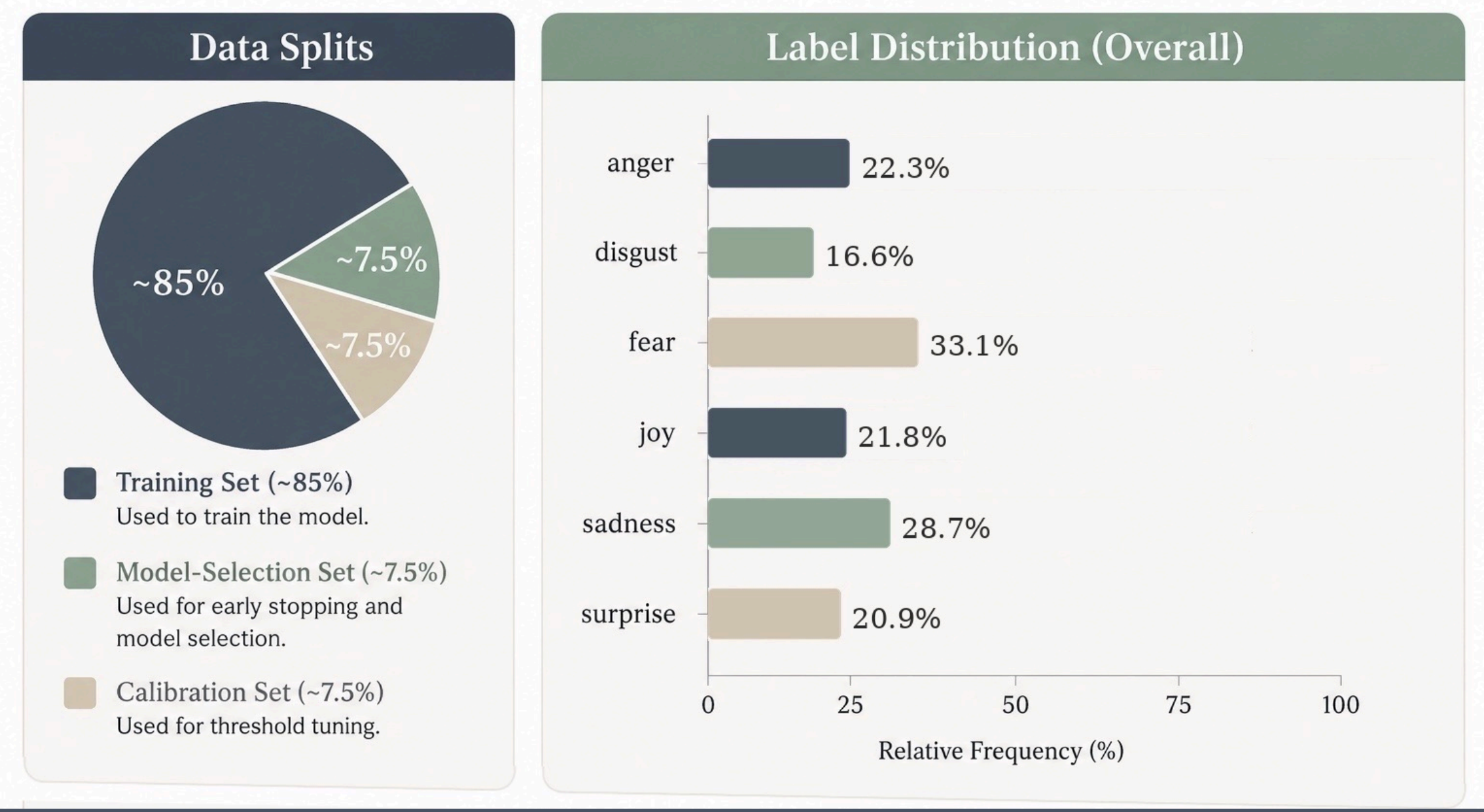
Approach

- Fine-tune XLM-RoBERTa-large for multi-label emotion prediction.
- Use focal loss and class weights to handle imbalanced rare labels.
- Apply language-balanced sampling so high-resource languages do not dominate training.
- Add translated silver data from English to German and Algerian Arabic using NLLB-200.
- Calibrate separate thresholds for each language and emotion to improve final predictions.



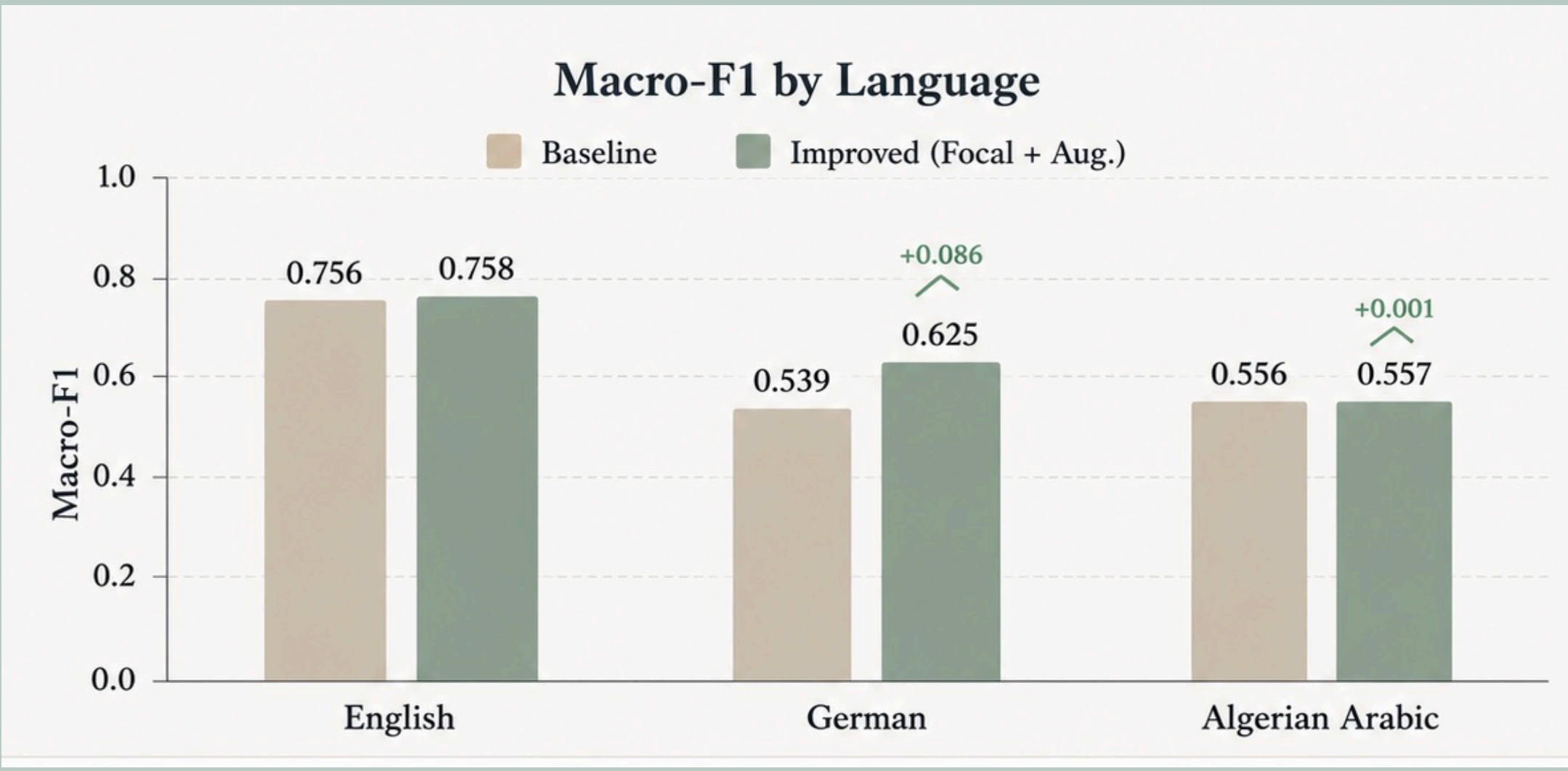
Data

- Dataset: BRIGHTER corpus from SemEval-2025 Task 11.
- Data type: short multilingual texts.
- Languages: English, German, and Algerian Arabic.
- Labels: anger, disgust, fear, joy, sadness, surprise.
- Task format: multi-label, each text can have multiple emotions.
- The official dev set was held out and scored once using per-language macro-F1.



Result

- The improved XLM-RoBERTa-large achieved the best result: 0.647 macro-F1.
- Performance increased from 0.617 to 0.647.
- The biggest gain was in German, from 0.539 to 0.625.
- English performed best, while German and Algerian Arabic were harder.
- Anger was easiest to predict; joy and surprise were hardest.
- Focal loss, augmentation, and threshold calibration improved rare labels.

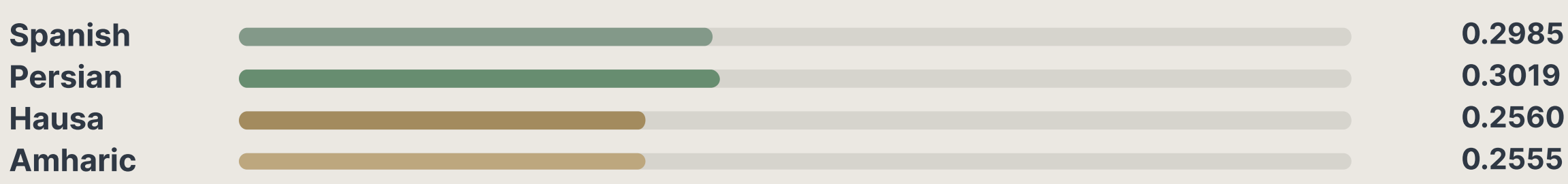


Conclusions

- Multilingual emotion detection remains challenging for low-resource languages.
- Handling rare emotion labels is key for stronger cross-lingual performance.
- Language-specific design choices improved model reliability.
- Smaller local models are promising, but need stronger evaluation.
- Multilingual transfer works, but still depends on data quality and label balance.

Unseen Language Evaluation

- Macro-F1 scores were measured on unseen languages: Spanish, Persian, Hausa and Amharic.



References

- [BRIGHTER Corpus / SemEval-2025 Task 11](#)
- [XLM-RoBERTa](#)
- [NLLB-200](#)
- [Focal Loss](#)